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Hardware-synchronized device for FAST-LIVO (Handheld & UAV).

[zhuanlan.zhihu.com/p/670136001](#)

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[Update] open-sourced the UAV platform equipped with hard-sync LIV se...

930219e · last week 🕒

📁 drone_cad	[Update] open-sourced the UAV platform equ...	last week
📁 handheld_cad	[fix] fix GPRMC time and checksum generatio...	10 months ago
📁 livox_ros_driver	[Fix] fix GPRMC timestamp mismatch with LiD...	5 months ago
📁 livox_ros_driver2	[Add] add modified livox_ros_driver2.	5 months ago
📁 mvs_ros_pkg	[Add] add BayerGB support.	5 months ago
📁 pics	[Update] open-sourced the UAV platform equ...	last week
📁 stm32_timersync-open	[fix] fix GPRMC time and checksum generatio...	10 months ago
📄 LICENSE	Initial commit	2 years ago
📄 README.md	[Update] open-sourced the UAV platform equ...	last week

LiDAR_Inertial_Visual_Handhold

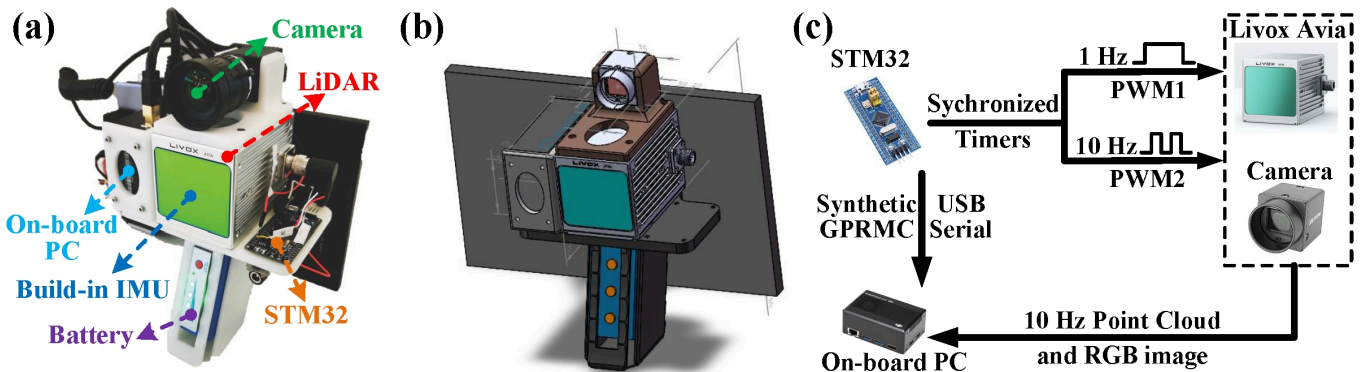
News

- **Mar. 24, 2025** - Open-sourced the UAV platform with hard-sync LIV sensors used in [FAST-LIVO2](#). 🎉
- **Oct. 27, 2024** - Add modified livox_ros_driver2 with time sync support for Mid360.
- **Oct. 11, 2024** - Add support for other pixel formats besides RGB8 (e.g., BayerRG8, BayerRG12Packed).
- **Oct. 03, 2024** - Add ARM support, fix the camera driver startup and shutdown bug, remove the SerialNumber dependency, resolve the auto-exposure mode switching issue, and add a TriggerEnable interface to allow testing the camera without the PPS signal.
- **Jul. 15, 2024** - Fix the bug related to LiDAR (IMU) timestamp compensation in the `livox_ros_driver`, remove the dynamic library dependency from `mvs_ros_driver`, and enable this synchronization scheme to use the official `livox_sdk`.
- **Jun. 28, 2024** - Update reproduction videos and some important notes.
- **May. 27, 2024** - Fix a bug in the calculation of the synthetic GPRMC timestamp, which could cause timestamp rollback. Additionally, we add checksum verification for GPRMC.
- **May. 20, 2024** - According to Livox Avia's pin requirements, we convert PPS TTL level to RS485 level and supplement it with more detailed electronic connection and material list.

📧 For further assistance or inquiries, please feel free to contact Chunran Zheng at zhengcr@connect.hku.hk.

1. Introduction

This repository provides the **CAD files** (with suffix **"*.SLDPRT** and ***.SLDASM"**) for **our handheld and uav-mounted devices**, which can be opened and edited with [Solidworks](#). All of the modules are suitable for printing with [FDM \(Fused Deposition Modeling\)](#). In addition, we have also open-sourced our **hardware synchronization scheme**, as well as the **STM32 source code** and **hardware wiring configuration instructions**.



2. Relevant guidance video

Thanks to the Bilibili uploader [GundaSmart](#) for making a detailed reproduction video. Everyone can follow the video to reproduce it. His sensor configuration includes a MID360 and a Hikvision pinhole camera.

[FAST-LIVO Reproduction \(Part 1\)](#)

[FAST-LIVO Reproduction \(Part 2\)](#)

[FAST-LIVO Reproduction \(Part 3\)](#)

[FAST-LIVO Reproduction \(Part 4\)](#)

[FAST-LIVO Reproduction \(Part 5\)](#)

3. Guide to installation

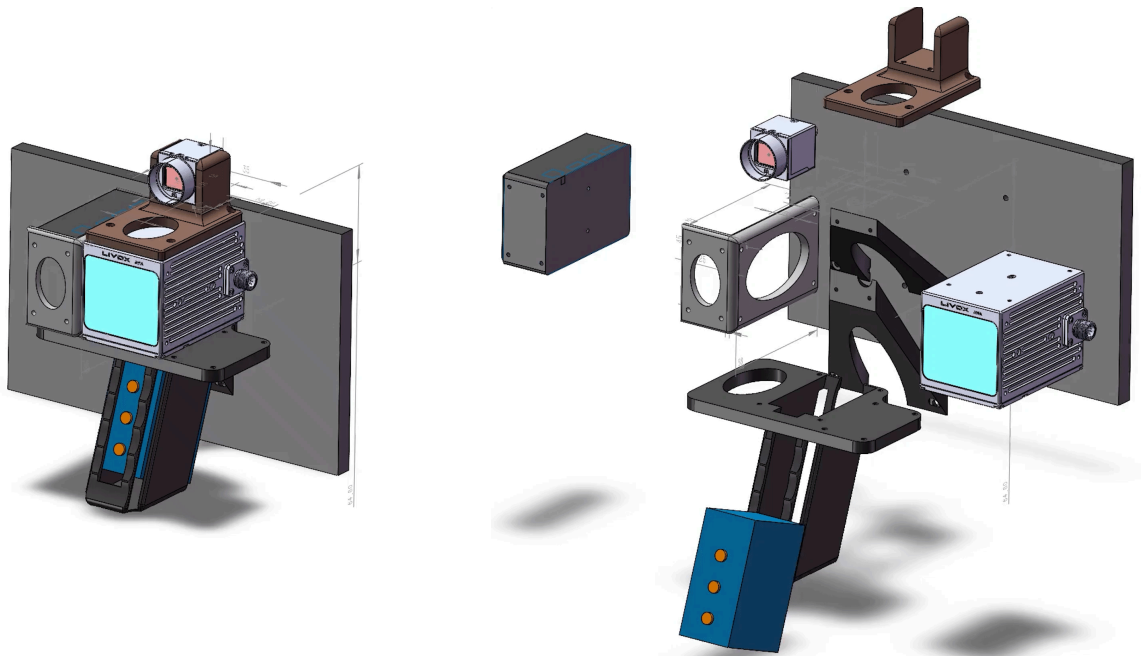
3.1 Root directory

```
|— handheld_cad/ - CAD source files for Handheld device
|   |— ...
|— drone_cad/ - CAD source files for UAV device
|   |— ...
|— livox_ros_driver/ - Livox LiDAR ROS driver
|   |— ...
|— mvs_ros_pkg/ - Camera driver
|   |— ...
|— stm32_timersync-open/ - Embedded engineering folder
|   |— USER/ - Main functionality folder
|   |— ...
|— README.md - Project homepage document
|— ...
```



3.2 Assembly instruction

The assembly instructions are demonstrated in the GIFs below. Each module is clearly labeled corresponding to the name of its STL file.



3.3 Electronic connection

The guide for the electronic connections is presented as follows:

Livox AVIA LiDAR M12	Peripheral Function	Diagram
PIN 1	Power: +	12P AVIA 12-Pin Interface, top: female bottom: male
PIN 2	Ground	
PIN 7	Ethernet: RX-	
PIN 6	Ethernet: RX+	
PIN 5	Ethernet: TX-	
PIN 4	Ethernet: TX+	
PIN 11 (Sync-)	RS485_Output B-	
PIN 12 (Sync+)	RS485_Output A+	

MVS Camera 6PIN	Name	I/O Type	Description	Peripheral Function	Diagram
PIN 1	DC_PWR	--	Power Supply		MVS Camera 6-Pin Interface
PIN 2	OPTO_IN	Line 0+	Optical Isolation Input	STM32 PA1	
PIN 3	GPIO	Line 2+	General Purpose Input/Output		
PIN 4	OPTO_OUT	Line 1+	Optical Isolation Output		
PIN 5	OPTO_GND	Line 0- / 1-	Optical Isolation Ground	STM32 GND	
PIN 6	GND	Line 2-	Ground		

STM32	Peripheral Function
PA1	MVS camera PIN2 (OPTO_IN)
PB5	RS485_Input (TXD)
PA9	TTL2USB (RXD)
VCC	RS485_Input (VCC) TTL2USB (VCC)

GND	RS485_Input (GND) TTL2USB (GND) MVS camera PIN5 (GND)	
RS485	Peripheral Function	
RS485_Output A+	LiDAR M12 PIN12 (Sync+)	
RS485_Output B-	LiDAR M12 PIN11 (Sync-)	
RS485_Output (GND)	LiDAR M12 PIN2 (GND)	

🔥 IMPORTANT:

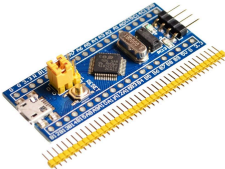
- STM32 PB5 (PPS signal) is converted from TTL to RS-485, resulting in RS485_Output A+ and RS485_Output B-;
- STM32 TXD (GPRMC) is converted from TTL to USB and sent to the PC;
- If you are using the Mid360, you can directly connect STM32 PB5 to LiDAR M12 pps interface;
- Before you `roslaunch livox_ros_driver livox_lidar_msg.launch`, you can use `sudo chmod a+rw /dev/ttyUSB0` grant permissions to the USB serial port;
- The default connection method in this repository directly connects to M12 without using the Livox converter. If you use the converter, you can connect STM32 PB5 (PPS signal) to the Livox converter Sync Port without converting TTL to 485 level. Refer to [Issue 19](#) for details.

4. How to run the driver

```
roslaunch mvs_ros_pkg mvs_camera_trigger.launch
roslaunch livox_ros_driver livox_lidar_msg.launch
rosbag record /livox/lidar /livox/imu left_camera/image
```



5. Bill of materials (only for reference)

Item	Pics	Purchasing list
Livox Avia LiDAR		Livox Avia
CMOS		MV-CA013-21UC
Camera Len		MVL-HF0628M-6MPE
STM32		STM32F103C8T6
Screen		IPS Screen 10"

Item	Pics	Purchasing list
Battery		4800mah
TTL to USB		TTL to USB
TTL to 485		TTL to 485

6. Alternative UAV Platform

For detailed equipment drawings, refer to the [drone_cad](#) folder.



Releases

No releases published

Packages

No packages published

Contributors 4

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Languages

● C 63.1% ● C++ 35.6% ● CMake 0.6% ● Assembly 0.4% ● Python 0.3% ● Shell 0.0%